jupyter Day3 Last Checkpoint: 4 hours ago

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Code JupyterLab Python [conda env:base] *

[5]:
```
!pip install pandas
```

```
Requirement already satisfied: pandas in c:\users\dell-\anaconda3\lib\site-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\dell-\anaconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\dell-\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\dell-\anaconda3\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\dell-\anaconda3\lib\site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\dell-\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Could not fetch URL https://pypi.org/simple/pip/: There was a problem confirming the ssl certificate: HTTPSConnectionPool(host='pypi.org', port=443):
Max retries exceeded with url: /simple/pip/ (Caused by SSLError(SSLCertVerificationError('A certificate chain processed, but terminated in a root cert
ificate which is not trusted by the trust provider.'))) - skipping
```

[18]:
```
import pandas as pd
a=pd.Series([1,2,3,4,5])
print(a) #it prints index,val and dt
print(a.values) #it print only val
print(a.index) #it print index
print(a.dtype) #it print dt

#float
b=pd.Series([1.2,2.3,3.5,4.5,5.6])
print(b) #it prints index,val and dt
print(b.values) #it print only val
print(b.index) #it print index
print(b.dtype) #it print dt

#boolean
c=pd.Series([True,False,True])
print(c) #it prints index,val and dt
print(c.values) #it print only val
print(c.index) #it print index
print(c.dtype) #it print dt
```

```
#all types
d=pd.Series([True,6.1])
print(d) #it prints index,val and dt
print(d.values) #it print only val
print(d.index) #it print index
print(d.dtype) #it print dt

e=pd.Series(["swetha","rithu","harshu","siva","saran","sam"])
print(e) #it prints index,val and dt
print(e.values) #it print only val
print(e.index) #it print index
print(e.dtype) #it print dt
print(e[1]) #accesing element
print(e.iloc[1]) #access element using loc
#slicing
print(e[1:3])
print(e.loc[1:3]) #it access the stop value with index val
print(e.iloc[2:5]) #it only shows the value before stop val
```

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
[1 2 3 4 5]
RangeIndex(start=0, stop=5, step=1)
int64
0    1.2
1    2.3
2    3.5
3    4.5
4    5.6
dtype: float64
[1.2 2.3 3.5 4.5 5.6]
RangeIndex(start=0, stop=5, step=1)
```

```
RangeIndex(start=0, stop=3, step=1)
float64
0     True
1    False
2     True
dtype: bool
[ True False  True]
RangeIndex(start=0, stop=3, step=1)
bool
0    True
1     6.1
dtype: object
[True 6.1]
RangeIndex(start=0, stop=2, step=1)
object
0    swetha
1     rithu
2    harshu
3      siva
4     saran
5       sam
dtype: object
['swetha' 'rithu' 'harshu' 'siva' 'saran' 'sam']
RangeIndex(start=0, stop=6, step=1)
object
rithu
rithu
1     rithu
2    harshu
dtype: object
1     rithu
2    harshu
3      siva
dtype: object
2    harshu
3      siva
4     saran
```

```
print(name.str.len()) #each word count
```

```
A    12
C    36
f    62
dtype: int64
A    12
B    24
C    36
D    48
E    50
f    62
dtype: int64
50
A    17
B    29
C    41
D    53
E    55
f    67
dtype: int64
24
12
232
18.40289832245635
62
12
38.666666666666664
42.0
C    36
D    48
E    50
f    62
dtype: int64
6
12     1
24     1
```

```
#we need to creating a indexing values,label used
import pandas as pd
arr=pd.Series([12,24,36,48,50,62],index=['A','B','C','D','E','f'])
print(arr[['A','C','f']])
print(arr)
print(arr['E']) #put quotes
#vectorized operation-no loop required
print(arr+5) #python list has looping but here there is no looping
print(arr.iloc[1]) #iloc is based on position but loc based on index values,iloc only allows int
print(arr.loc['A'])

#statistical function
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())

#numpy only have percentile but pandas not

#boolean indexing
print(arr[arr>30]) #print with labeled values

#count values
print(len(arr)) #all val count

#it return individual value count
print(arr.value_counts()) #12 time how many time etc...

#string operation
name=pd.Series(['jack','vishnu','anitha','sashmitha'])
```

```
36    1
48    1
50    1
62    1
Name: count, dtype: int64
0         JACK
1       VISHNU
2       ANITHA
3    SASHMITHA
dtype: object
0    4
1    6
2    6
3    9
dtype: int64
```

```python
import pandas as pd
a=pd.Series({"gayathri":16,
             "rithanyaa":38,
             "bavanthika":8}) #only 8 not 08
print(a)
print(a["rithanyaa"])
```

```
gayathri      16
rithanyaa     38
bavanthika     8
dtype: int64
38
```

```python
#index alignment
s1=pd.Series([10,20,30],['A','B','C'])
s2=pd.Series([1,2,3],['B','C','D'])
print(s1+s2)  #if 1st have A 2nd have no A(empty) it prints nan and b,c in both so add),returns in float bcz nan is float if 1 one convert all into fl
s=pd.Series([1,2,3,4,5])
print(s.count())
```

```
print(s.count())
print(s.unique()) #no repetition allowed
print(s.nunique()) #unique values count
```

```
A     NaN
B    21.0
C    32.0
D     NaN
dtype: float64
5
[1 2 3 4 5]
5
```

```
import pandas as pd
#method1:from dictionary
data={
    'name':['monika','nagul','prem'],
    "rollno":[27,20,24],
    'Age':[19,20,21]}
df=pd.DataFrame(data) #convert data intp df
print(df)

#from series to dataframe8
s1=pd.Series([1,2,3,4,5])
s2=pd.Series([5.6,9.4,8.7,5.9,7.8])
s3=pd.Series([True,False,True,False])
df=pd.DataFrame({'sno':s1,
                 'cgpa':s2,
                 'attendance':s3})
print(df)

#list of list
data=[
```

```python
    ['athish',22,98],
    ['rithika',20,89],
    ['pratheksha',20,97]]
df=pd.DataFrame(data) #it shows with matrix 0,1,2
print(df)

dat=[
    ['athish',22,98],
    ['rithika',20,89],
    ['pratheksha',20,97]]
df=pd.DataFrame(dat,columns=['name','age','mark']) #it shows with col name
print(df)

#list of dictionary
data=[
    {"name":"sakthi","course":"data science","grade":'A'},
    {"name":"madhu","course":"aiml","grade":'A+'},
    {"name":"ragavi","course":"data science","grade":'A++'}]
df=pd.DataFrame(data)
print(df) #index 0,1,2

data=[
    {"name":"sakthi","course":"data science","grade":'A'},
    {"name":"madhu","course":"aiml","grade":'A+'},
    {"name":"ragavi","course":"data science","grade":'A++'}]
df=pd.DataFrame(data,index=["std1","std2","std3"])
print(df) #index changed std1 etc
#adding one extra col
df['marks']=[98,87,67]
print(df)
#add one extra row
std4=pd.DataFrame([{"name":"ragu","course":"data science","grade":'B',"marks":98}],index=["std4"])
df=pd.concat([df,std4])
print(df)
```

```python
#delete value
df=df.drop('grade',axis=1) #dlt col
print(df)

#another method
df.drop('std3',axis=0,inplace=True) #inplace means changes 6made in the original frame directly
print(df)

#sorting
df.sort_values('marks',inplace=True)   #asc default
print(df)

df.sort_values('marks',ascending=False,inplace=True) #get des put asc false
print(df)

#statistical func for particular col
print(df['marks'].max())
```

```
     name  rollno  Age
0  monika      27   19
1   nagul      20   20
2    prem      24   21
 sno  cgpa attendance
   1   5.6       True
   2   9.4      False
   3   8.7       True
   4   5.9      False
   5   7.8        NaN
            0   1   2
     athish  22  98
    rithika  20  89
 pratheksha  20  97
```

```
#create multiple rows
students=pd.DataFrame([{"name":"sanjay","course":"csda","grade":'C',"marks":77},
                       {"name":"divya","course":"ds","grade":'A',"marks":82}],index=["std5","std6"])
df=pd.concat([df,students])
print(df)

#column access by bracket notation and dot notation
print(df.course)
print(df[['grade','marks']])
print(df[['grade']])

#shape attributes
print(df.shape) #how many rows and cols

#find no of rows
print(df.shape[0])
#find no of cols
print(df.shape[1])

#Dtype attribute
#used to check data type of all columns
print(df.dtypes)
#info
print(df.info())
print(df.describe()) #print count,mean,min etc....
#select specific row
print(df.loc[['std1','std4']])
print(df.iloc[3])

#updatevalue
df.loc['std3','grade']='B++'
print(df)
```

```
2  pratheksha   20    97
         name  age  mark
0      athish   22    98
1     rithika   20    89
2  pratheksha   20    97
     name        course grade
0  sakthi  data science     A
1   madhu          aiml    A+
2  ragavi  data science   A++
        name        course grade
std1  sakthi  data science     A
std2   madhu          aiml    A+
std3  ragavi  data science   A++
        name        course grade  marks
std1  sakthi  data science     A     98
std2   madhu          aiml    A+     87
std3  ragavi  data science   A++     67
        name        course grade  marks
std1  sakthi  data science     A     98
std2   madhu          aiml    A+     87
std3  ragavi  data science   A++     67
std4    ragu  data science     B     98
        name        course grade  marks
std1  sakthi  data science     A     98
std2   madhu          aiml    A+     87
std3  ragavi  data science   A++     67
std4    ragu  data science     B     98
std5  sanjay          csda     C     77
std6   divya            ds     A     82
std1    data science
std2            aiml
std3    data science
std4    data science
std5            csda
std6              ds
Name: course, dtype: object
     grade  marks
```

```
      grade  marks
std1      A     98
std2     A+     87
std3    A++     67
std4      B     98
std5      C     77
std6      A     82
      grade
std1      A
std2     A+
std3    A++
std4      B
std5      C
std6      A
(6, 4)
6
4
name      object
course    object
grade     object
marks      int64
dtype: object
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, std1 to std6
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    6 non-null      object
 1   course  6 non-null      object
 2   grade   6 non-null      object
 3   marks   6 non-null      int64
dtypes: int64(1), object(3)
memory usage: 240.0+ bytes
None
           marks
count   6.000000
mean   84.833333
```

```
import pandas as pd
#read csv file
df=pd.read_csv('customers-100.csv',usecols=['First
print(df)
#Read only 1st 5 data
print(df.head())
#last 5 data
print(df.tail())
print(df.to_string())
print(df['Country']) #specific col
```

```
   First Name                     Country
0      Sheryl                       Chile
1     Preston                    Djibouti
2         Roy         Antigua and Barbuda
3       Linda          Dominican Republic
4      Joanna  Slovakia (Slovak Republic)
..        ...                         ...
95       Karl                      Guyana
96       Lynn                   Sri Lanka
97       Fred             Solomon Islands
98     Yvonne                       Aruba
99   Clarence                    Honduras

[100 rows x 2 columns]
  First Name                     Country
0     Sheryl                       Chile
1    Preston                    Djibouti
2        Roy         Antigua and Barbuda
3      Linda          Dominican Republic
4     Joanna  Slovakia (Slovak Republic)
   First Name          Country
95       Karl           Guyana
96       Lynn        Sri Lanka
```

		Aruba
99	Clarence	Honduras
	First Name	
0	Sheryl	Country
1	Preston	Chile
2	Roy	Djibouti
3	Linda	Antigua and Barbuda
4	Joanna	Dominican Republic
5	Aimee	Slovakia (Slovak Republic)
6	Darren	Bosnia and Herzegovina
7	Brett	Pitcairn Islands
8	Sheryl	Bulgaria
9	Michelle	Cyprus
10	Carl	Timor-Leste
11	Jenna	Guernsey
12	Tracey	Vietnam
13	Kristine	Togo
14	Faith	Sri Lanka
15	Miranda	Singapore
16	Caroline	Oman
17	Greg	Western Sahara
18	Clifford	South Georgia and the South Sandwich Islands
19	Joanna	French Polynesia
20	Maxwell	Malta
21	Kiara	Netherlands
22	Colleen	Paraguay
23	Janet	Lao People's Democratic Republic
24	Shane	Albania
25	Marcus	Panama
26	Dakota	Belarus
27	Frederick	Switzerland
28	Stefanie	Saint Vincent and the Grenadines
29	Kent	Tanzania
30	Jack	Zimbabwe
31	Tom	Denmark

32	Gabriel	Liechtenstein
33	Kaitlyn	United States of America
34	Faith	Bahamas
35	Tammie	Belize
36	Nicholas	Uruguay
37	Jordan	Solomon Islands
38	Bruce	Montenegro
39	Sherry	Poland
40	Natalie	Dominican Republic
41	Bryan	Burkina Faso
42	Wayne	Bolivia
43	Luis	Bulgaria
44	Rhonda	Monaco
45	Joanne	Palau
46	Geoffrey	Uzbekistan
47	Gloria	Ghana
48	Brady	United Arab Emirates
49	Latoya	Belarus
50	Gerald	Canada
51	Samuel	Algeria
52	Patricia	Swaziland
53	Stacie	Madagascar
54	Robin	Ecuador
55	Ralph	Palestinian Territory
56	Phyllis	Saint Barthelemy
57	Danny	United Arab Emirates
58	Kathy	South Georgia and the South Sandwich Islands
59	Kelli	Sao Tome and Principe
60	Lynn	Portugal
61	Shelley	Togo
62	Eddie	Ethiopia
63	Chloe	Netherlands
64	Eileen	Liberia
65	Fernando	Lithuania
66	Makayla	New Caledonia
67	Tom	Kiribati
68	Virginia	French Southern Territories

```
86       Duane                                                Maldives
87      Alison                                                   Benin
88      Vernon                                                   Yemen
89        Lori                                                 Namibia
90        Nina                                                  Bhutan
91       Shane                                                 Hungary
92      Collin                                                Anguilla
93      Sherry                                         Solomon Islands
94     Darrell                                                    Mali
95        Karl                                                  Guyana
96        Lynn                                               Sri Lanka
97        Fred                                         Solomon Islands
98      Yvonne                                                   Aruba
99    Clarence                                                Honduras
0                          Chile
1                       Djibouti
2            Antigua and Barbuda
3             Dominican Republic
4     Slovakia (Slovak Republic)
                   ...
95                        Guyana
96                     Sri Lanka
97               Solomon Islands
98                         Aruba
99                      Honduras
Name: Country, Length: 100, dtype: object
```

69	Riley	Canada
70	Alexandria	Iran
71	Richard	Morocco
72	Candice	Zimbabwe
73	Anita	Russian Federation
74	Regina	Solomon Islands
75	Debra	United States Virgin Islands
76	Brittany	Kyrgyz Republic
77	Cassidy	Myanmar
78	Laurie	Dominica
79	Alejandro	Iceland
80	Leslie	Micronesia
81	Kathleen	Saint Vincent and the Grenadines
82	Hunter	Isle of Man
83	Chad	Oman
84	Corey	Fiji
85	Emma	New Zealand
86	Duane	Maldives
87	Alison	Benin
88	Vernon	Yemen
89	Lori	Namibia
90	Nina	Bhutan
91	Shane	Hungary
92	Collin	Anguilla
93	Sherry	Solomon Islands
94	Darrell	Mali
95	Karl	Guyana
96	Lynn	Sri Lanka
97	Fred	Solomon Islands
98	Yvonne	Aruba
99	Clarence	Honduras

0	Chile
1	Djibouti
2	Antigua and Barbuda
3	Dominican Republic
4	Slovakia (Slovak Republic)